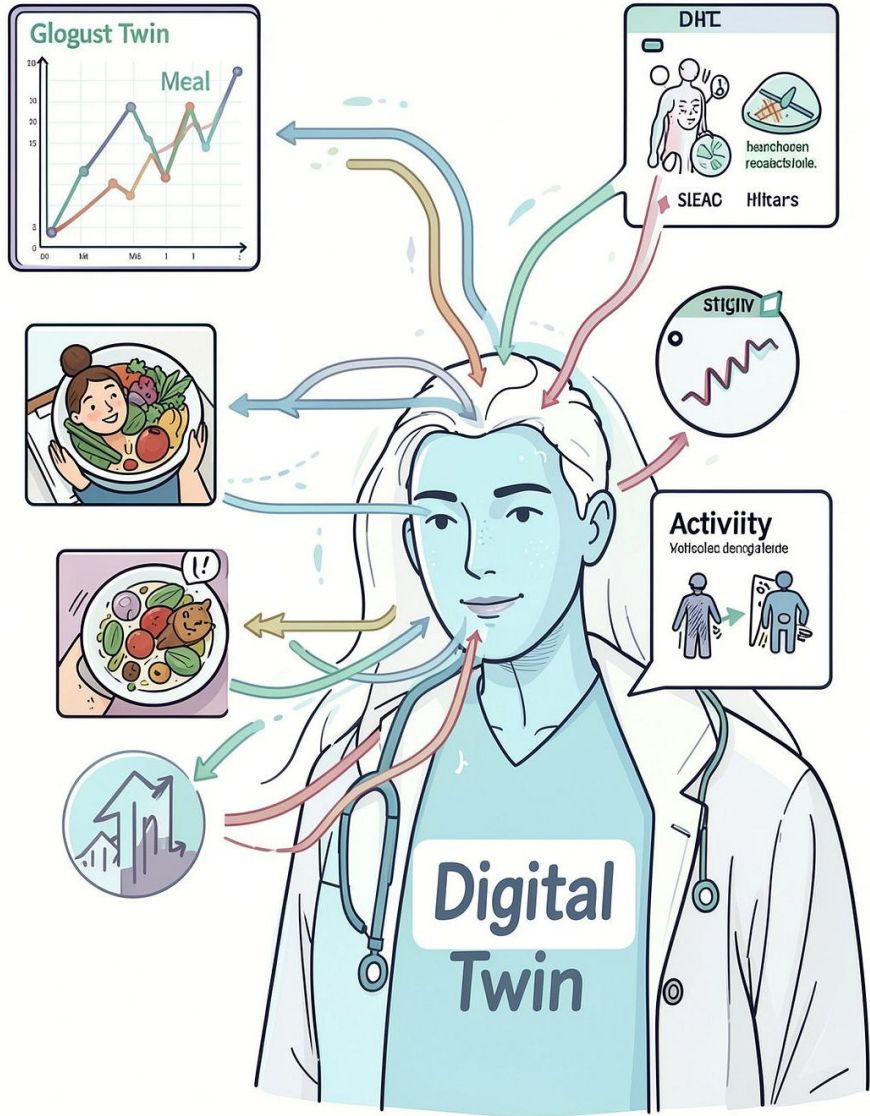




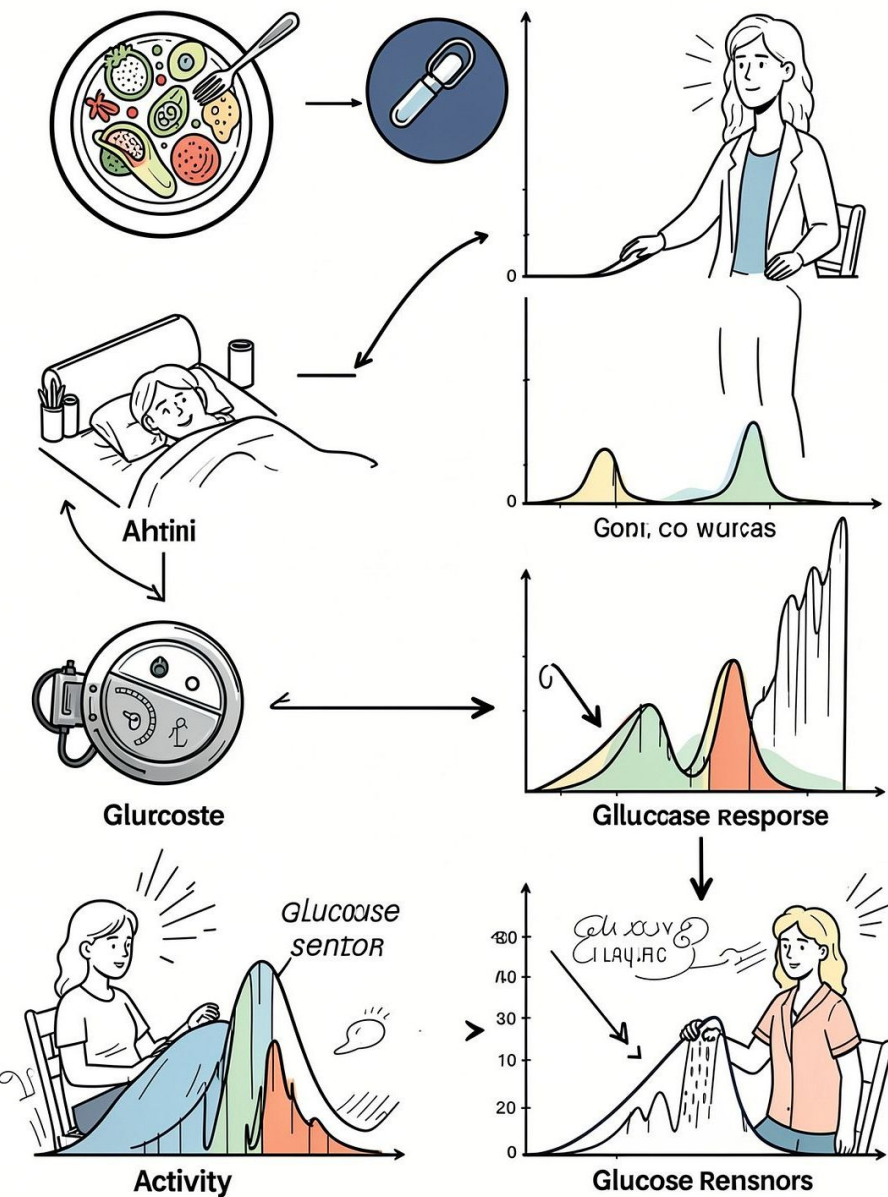
GlycoTwin

A Personalized Digital Twin for Glucose Management

Happiest Health Digital Twin Challenge 2026

Team: GlucoStudio • **College :** VIT Bhopal University

Team members: Abhi Pandey ,Tanishq Das

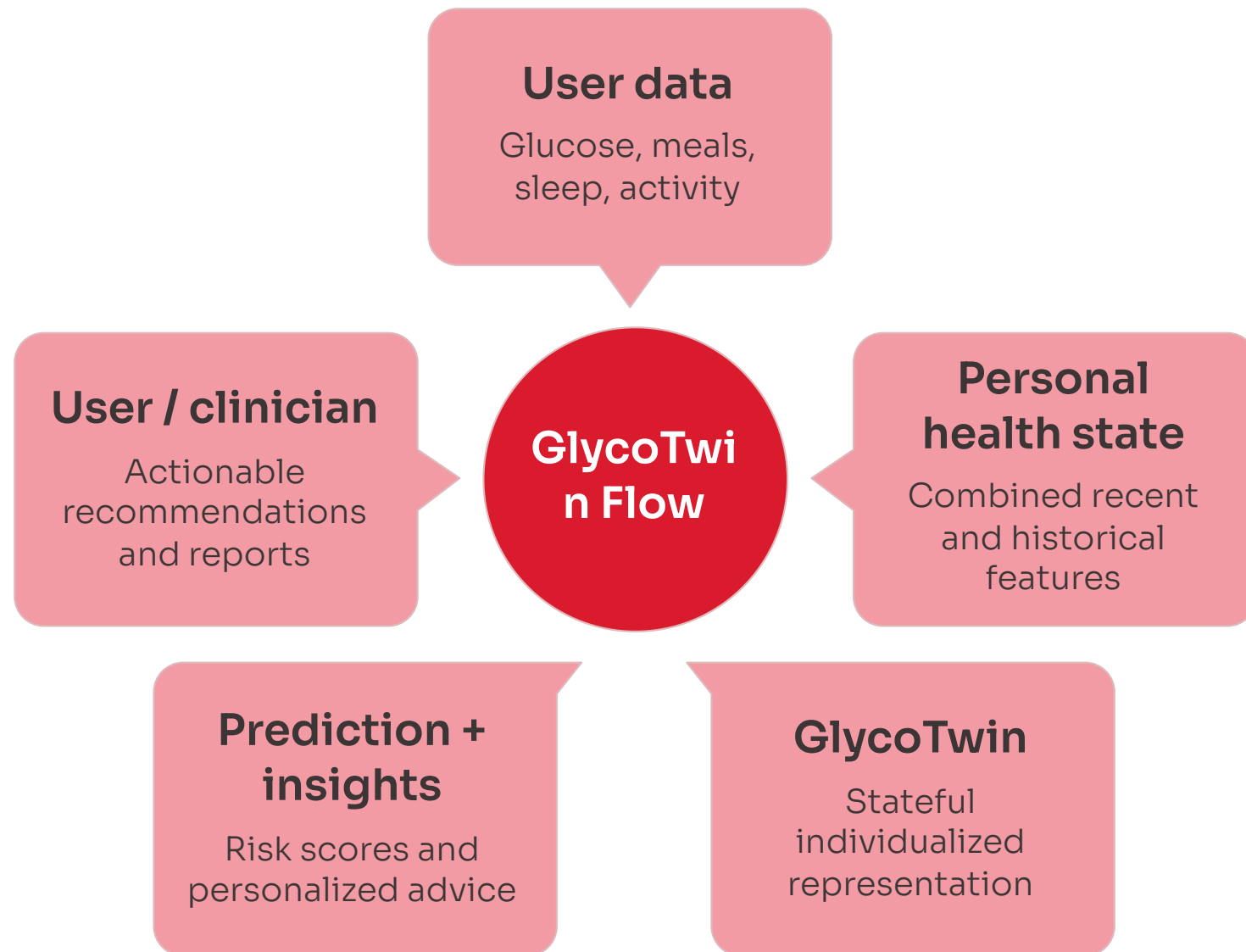
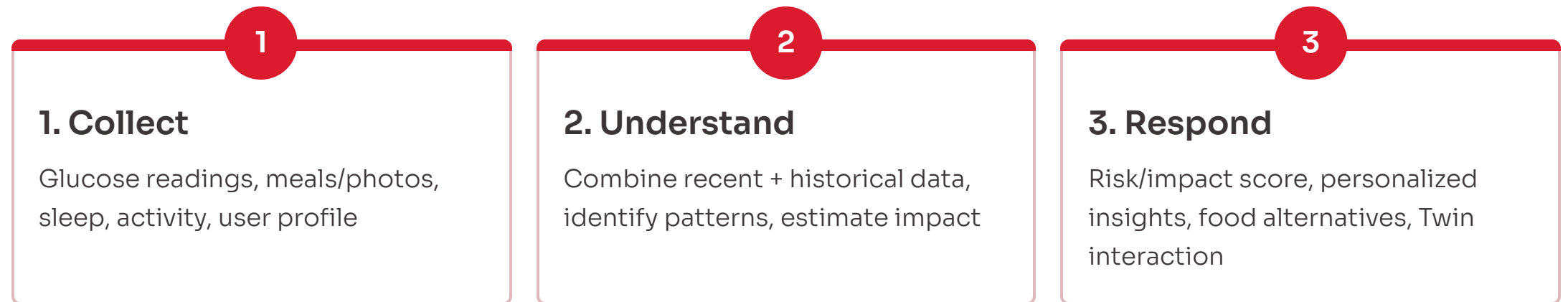


The Problem We Started With

- Daily decisions are repetitive: "What will this meal do to my glucose?"
- Responses depend on recent activity, sleep and prior meals — not just the food.
- Generic guidance misses individual patterns and context.

Main message: Glucose response is personal — the same meal can affect people differently.

Our Approach: GlycoTwin

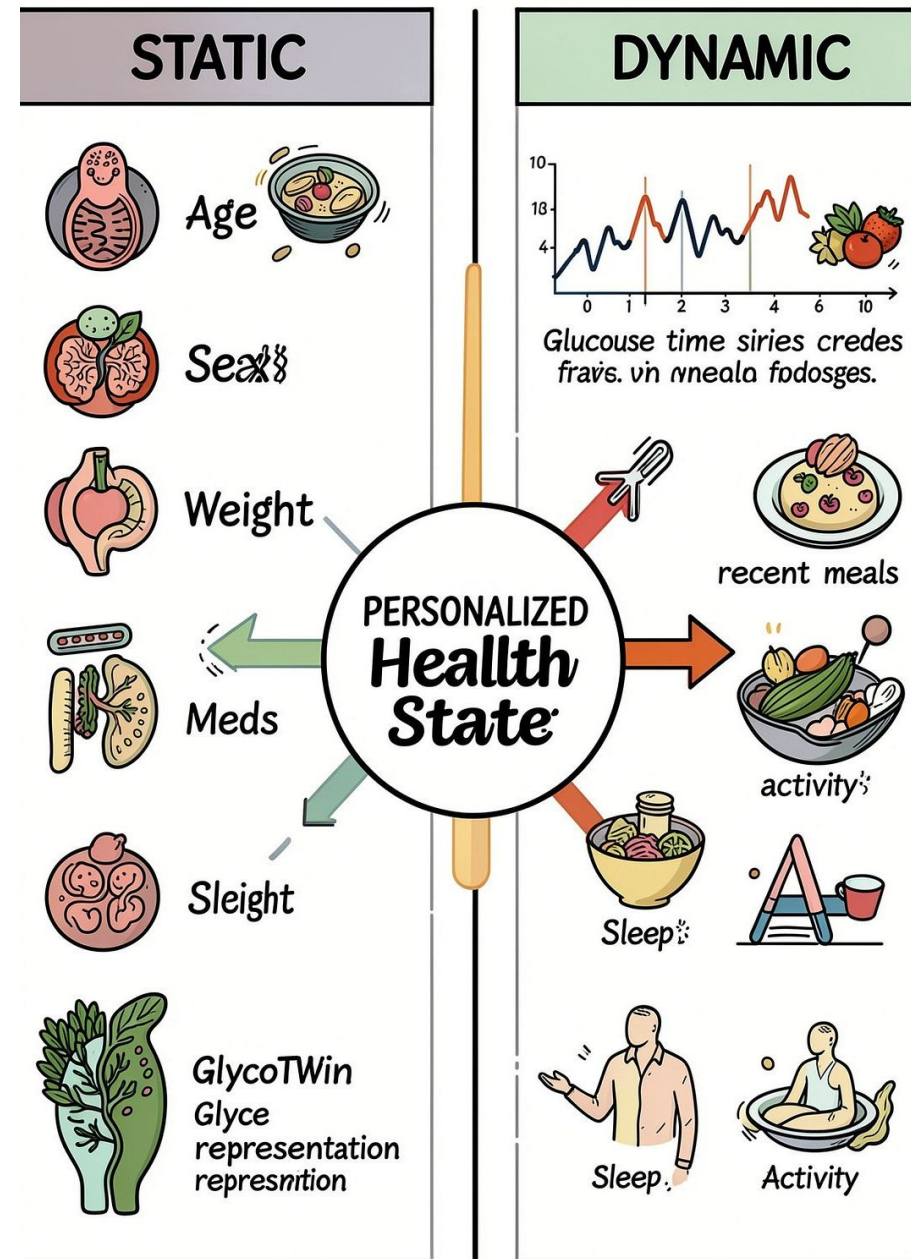


From Health Data to a Digital Twin

GlycoTwin maintains a changing representation of the individual by combining:

- **Static:** profile and personal parameters
- **Dynamic:** glucose history, meal history, sleep, activity, recent observations

New data updates the representation so insights reflect the person's current context, not only population averages.



System Workflow



Key flow in 7 seconds: data collected on device → synced to backend → processed by scoring logic → Twin updated → user sees contextual insight.

Technical Architecture



Frontend

React Native ▪ Expo ▪ TypeScript / JavaScript



Backend

Node.js ▪ Express ▪ REST APIs



Database

MongoDB (user, event, nutrition collections)



AI & Tools

Google Gemini (text/image assistance where used)



Integrations

Apple Health / HealthKit (where implemented), Barcode scanning, Open Food Facts, nutrition APIs



Prediction

GlycoTwin prediction / scoring logic (heuristic & feature-based)

How We Estimate Glucose Impact

We implemented a scoring/prediction engine that combines meal features and personal context. It is described in the repository as rule-based / heuristic logic rather than a clinically validated ML model.

01

Inputs

Meal (carbs, fiber, protein, category), recent glucose trend, activity, sleep, profile

02

Processing

Feature extraction → rule-based weights and heuristics → scoring function

03

Output

Glucose impact / spike risk (interpretable score used for insight generation)

📄 Repository notes: if the code returns a 0–100 style score, we label it as a heuristic risk score. No clinical validation numbers are reported.

From a Meal Photo to Personalized Insight



1. User captures meal photo or scans barcode
2. System tries barcode → Open Food Facts / nutrition API lookup
3. If no barcode, image-assisted identification using Gemini or metadata (where implemented)
4. Nutrition values combined with user's recent context (glucose, activity, sleep)
5. GlycoTwin scoring engine estimates impact
6. App shows insight: expected impact, suggestions, or alternatives

Note: Barcode lookup path and Open Food Facts integration are implemented; AI fallback is used in parts of the prototype where available.

Meet the Twin

Screenshots show the prototype UI and how the Twin aggregates recent data into an actionable view.



The Twin uses the user's own history to contextualize insights rather than only generic rules.

Why Personalization Matters

Person A

Good sleep ▪ Higher recent activity ▪ Glucose trending stable

Person B

Poor sleep ▪ Low activity ▪ Recent elevated readings

Same meal, different context → different predicted impact and tailored insight. Our focus is contextual interpretation, not definitive medical advice.

Generic Food Info

Personal Context

GlycoTwin

Relevant Insight